
RESEARCH SEMINAR · WORK IN PROGRESS

A recursive-dynamic CGE model of Burundi

Dual exchange rates, foreign-exchange rationing of fuel, and what they mean for reform

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Code and data: github.com/kodzovee9/burundi-cge

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The findings, interpretations and conclusions expressed here are those of the author and do not necessarily represent the views of the World Bank, its Executive Directors or the governments they represent.

MOTIVATION

Burundi's binding constraint is foreign exchange

1.58 →
2.17

parallel / official exchange
rate, 2019 → 2024

≈ 70 %

of imports priced at the
parallel rate (BRB)

−24 %

official fuel imports, 2024
against 2019 (BRB
tonnage)

3.35 ×

black-market fuel price
over the pump price, 2024

The model has to answer: what does unifying the exchange rate buy? How should scarce official foreign exchange be allocated? What do public investment, human development and mining add on top?

OUTLINE

Six parts, ending with the questions for you

- 1 Lineage and data** GEM-Core, what is added, the 2019 SAM
 - 2 Building blocks** twelve blocks with their equations; closures; solution method
 - 3 Transmission channels** unification, FX and fuel, fiscal, public investment, human development, mining, external shocks
 - 4 Validation** against the GAMS original, and against Burundi 2019–2024
 - 5 Results and simulations** reform scenarios 2026–40, pass-through sensitivity, what could be run next
 - 6 Questions for feedback** six modelling choices the data do not settle
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Built on GEM-Core, re-implemented and extended

From GEM-Core (World Bank)

- Nested CES production, Leontief intermediates
- Eight labour types on a wage curve
- Armington imports, CET exports
- Dual exchange rate with a premium rent
- Quotas on fuel and chemicals
- Recursive dynamics; public capital raises TFP

Added for Burundi

- Fuel quota set by official foreign exchange
- Informal fuel from the DRC and Tanzania
- Firms trade fuel against value added
- Households switch to firewood and charcoal
- Premium pass-through θ ; rent partly a real cost
- Idle-resource closure for mining

Written in Python and solved as a square system with Newton's method. It reproduces the GAMS/PATH reference application, and one switch removes every addition.

A 2019 SAM consistent with INSBU's rebased accounts

Accounts	Detail
Activities	19, agriculture to other services
Commodities	20, incl. refined petroleum (all imported)
Labour	8: male / female × 4 education levels
Other factors	private and public capital, land, mining resource
Institutions	rural and urban households, government, rest of world

2019 structure	Value
Private / government consumption, % GDP	90.3 / 9.9
Investment, % GDP	11.7
Exports / imports, % GDP	5.3 / 18.4
Current account, % GDP	−7.2
Agriculture, % of value added	41.9
Labour, % of factor income	38.2
Imports at the official rate, %	28.6

The 28.6 % of imports at the official rate (fuel and chemicals) matches BRB's $\approx$ 30 %. Elasticities are standard GEM-Core values, not Burundi estimates.

2 · BUILDING BLOCKS

Twelve building blocks, one equilibrium per year

1 · Production

output, factor and input demand
· A.1–A.8

2 · Fuel substitution

fuel vs value added per unit of
output · A.9–A.10

3 · Factor markets

wages, unemployment, rents ·
A.11–A.13

4 · Trade

imports, exports, domestic sales
· A.14–A.15

5 · Dual exchange rate

trade prices, premium rent, its
real cost · A.16–A.19, A.5a

6 · Quotas and FX rationing

official fuel imports, quota rent ·
A.20–A.21

7 · Informal fuel

smuggled volume and price ·
A.22–A.23

8 · Households

income, saving, consumption ·
A.24–A.26

9 · Government

revenue, spending, deficit, debt ·
A.27–A.28

10 · Saving, investment, BoP

investment, foreign saving ·
A.29–A.32

11 · Clearing and prices

commodity balances, CPI,
numeraire, GDP · A.3.11

12 · Dynamics

capital, labour, debt between
years · A.33–A.34

Notation. Upper case: endogenous variables. Lower case and Greek: parameters. Superscript 0: base year. Indices: a activities (19), c commodities (20), f factors, h households (2), t years. Blocks marked in orange are added to GEM-Core.

BLOCK 1 • PRODUCTION

Nested CES value added, Leontief inputs

$$QA_a \text{ } VXI_a = TFP_a \phi_a \left[\sum_f \delta_{f,a} \left(FPRD_{f,a} \text{ } QF_{f,a} \right)^{-\rho_a} \right]^{-1/\rho_a} \tag{A.1}$$

$$WFA_{f,a} = PVA_a \frac{\partial (QA_a \text{ } VXI_a)}{\partial QF_{f,a}}, \qquad WFA_{f,a} = WF_f \text{ } WFDIST_{f,a} (1 + tfa_{f,a}) \tag{A.2–A.3}$$

$$QINT_{c,a} = ica_{c,a} \text{ } QA_a, \qquad QF_{r,a} = ifa_{r,a} \text{ } QA_a \text{ } VXI_a \tag{A.7, A.4}$$

$$TFP_{a,t} = tfp_{a,t}^{ex} \left(1 + TFPSCAL_t \text{ } tfp01_a \right) \left(\frac{TRDGDP_t}{TRDGDP^0} \right)^{\epsilon_a} RW_t \tag{A.5}$$

QA activity output	VXI value added per unit of output (block 2)	QF factor use; FPRD factor productivity
φ, δ, ρ CES shift, shares, exponent; σ = 1/(1+ρ) = 1.25	PVA value-added price	WFA, WF, WFDIST sector factor price, economy-wide price, sector differential
ica, ifa input and resource coefficients; r = mining resource	TFPSCAL scalar backed out in calibration	TRDGDP trade over GDP; ε = 0.01; RW rent-seeking cost (block 5b)

Labour inside (A.1) is itself CES: across education levels σ = 4, male/female within a level σ = 2

Firms can trade fuel against value added

$$\frac{FXI_a}{VXI_a} = \left[\frac{PQD_{fuel,a} / PQD_{fuel,a}^0}{PVAC_a / PVAC_a^0} \right]^{-\sigma^F} \tag{A.9}$$
$$(1 - s_a^F) VXI_a^{\rho^F} + s_a^F FXI_a^{\rho^F} = 1, \quad \rho^F = \frac{\sigma^F - 1}{\sigma^F} \tag{A.10}$$

- FXI** fuel per unit of output
- VXI** value added per unit of output
- PQD fuel,a** fuel purchase price
- PVAC** unit cost of value added plus the resource
- s F** 2019 fuel share of (value added + fuel): electricity 0.45, transport 0.31, construction 0.20
- σ F = 0.1** elasticity of substitution

Reading. Both indices equal 1 at base prices, so the base year is unchanged. A 10 % rise in fuel's relative price cuts fuel per unit of output by about 1 % and raises value added per unit of output: saving fuel costs factors. $\sigma^F \rightarrow 0$ returns GEM-Core's Leontief technology, under which the 2024 fuel cut has no solution.

A wage curve for labour, fixed capital within a year

$$QFS_f (1 - UERAT_f) = \sum_a QF_{f,a} \quad (A.11)$$

$$\frac{WF_f}{CPI} = \frac{WF_f^0 fpr d_{f,t}}{CPI^0} \left(\frac{UERAT_f}{UERAT_f^0} \right)^{\eta^{wf}} \quad (A.12)$$

$QF_{k,a,t}$ predetermined, $WFDIST_{k,a}$ clears each sector

$$0 \leq UERAT_r \perp WF_r \geq WF_r^{res} \quad (A.13)$$

QFS labour supply, 8 types (sex × education)

UERAT unemployment rate (0.3–10.5 % in 2019)

WF, CPI wage, consumer price index

fprd labour-productivity index

$\eta^{wf} = -0.5$ wage-curve elasticity

k, r capital, mining resource

WF res reservation rent: the reference path

$\perp$ complementarity: one of the two inequalities binds

Reading. Labour demand moves wages and unemployment together; land and capital earn sector-specific rents. The idle-resource option lets a larger mining endowment stay in the ground rather than drive its rent to zero.

Armington imports, CET exports, small open economy

$$QQ_c = \phi_c^q [\delta_c^m QM_c^{-\rho_c^q} + \delta_c^d QD_c^{-\rho_c^q}]^{-1/\rho_c^q}, \quad \frac{QM_c}{QD_c} = \left[\frac{PDD_c}{PM_c} \frac{\delta_c^m}{\delta_c^d} \right]^{\sigma_c^q} \tag{A.14}$$

$$QX_c = \phi_c^x [\delta_c^e QE_c^{\rho_c^x} + \delta_c^s QD_c^{\rho_c^x}]^{1/\rho_c^x}, \quad \frac{QE_c}{QD_c} = \left[\frac{PE_c}{PDS_c} \frac{\delta_c^s}{\delta_c^e} \right]^{\sigma_c^x} \tag{A.15}$$

QQ composite supply to the domestic market

PM, PDD import and domestic demand prices

$\sigma_q = 1/(1+\rho_q)$ Armington elasticity

QM, QD imports, domestic sales

PE, PDS export and domestic supply prices

$\sigma_x = 1/(\rho_x - 1)$ CET elasticity

QX, QE domestic output, exports

φ, δ shift and share parameters

Both: 2.0 primary, 1.5 manufactures, 0.9 services

Reading. World prices PWM, PWE are exogenous in US dollars. Import shares of supply: fuel 100 %, textiles and other manufactures 74 %; export shares of output: mining 67 %, most others under 10 %. The CET elasticity governs how far a real depreciation raises exports.

Two exchange rates, one premium, one rent

$$PM_c = (1 + tm_c + PRQMBAR_c) PWM_c EXR [(1 - shrom_c) PREXR^m + shrom_c] + \text{margins} \tag{A.16}$$

$$PE_c = (1 - te_c) PWE_c EXR [(1 - shroe_c) PREXR + shroe_c] - \text{margins} \tag{A.17}$$

$$PREXR_t^m = P_t^{anc} + \theta (PREXR_t - P_t^{anc}) \tag{A.18}$$

$$YPREXRT_c = (1 - shrom_c)(PREXR^m - 1) EXR PWM_c QM_c - (1 - shroe_c)(PREXR - 1) EXR PWE_c QE_c \tag{A.19}$$

EXR official exchange rate, BIF per USD	PREXR parallel ÷ official rate: 1.58 (2019), 2.17 (2024)	PREXR m premium that reaches import costs
θ pass-through (added); θ = 1 is GEM-Core	P anc anchor: reference path, or 2019 in the backcast	shrom, shroe shares at the official rate: fuel, chemicals 1; other imports 0; coffee, tea receipts 0.75
PWM, PWE world prices, USD	tm, te tariff, export tax; PRQMBAR quota rent rate	YPREXRT premium rent, credited to private capital

Part of the rent is competed away in rent-seeking

$$RW_t = \frac{1 - \omega RSH_t^{eff}}{1 - \omega RSH_{2019}} \quad (A.5a)$$

$$RSH_t^{eff} = (1 - \lambda) RSH_{t-1}^{eff} + \lambda \frac{\sum_c YPREXRT_{c,t}}{GDPMP_t} \quad (A.5a)$$

RW multiplies TFP in every activity (A.5)

$\omega = 0.25$ share of the rent that is a real cost; 0 and 0.5 as sensitivities; 0 = GEM-Core

RSH eff effective rent share of GDP

$\lambda = 1/3$ share of the gap closed each year

YPREXRT, GDPMP premium rent (A.19); nominal GDP

Reading. In GEM-Core the rent is a pure transfer, so ending it only redistributes income. Following Krueger (1974), resources spent queuing for official FX, on informal intermediation and on lobbying produce nothing. At the 2.5 premium the rent is 17 % of GDP (2019), rising to 21 % (2040); $\omega = 0.25$ means about 4–5 % of GDP is used up. $RW = 1$ in 2019, so the base year is unchanged.

The fuel quota is BRB's foreign-exchange allocation

$$QMBAR_{c,t} - QM_c \geq 0 \perp PRQMBAR_c \geq 0 \tag{A.20}$$

$$YPRQMBART_c = PRQMBAR_c PWM_c EXR QM_c$$

$$QMBAR_{fuel,t} = \varphi_t \frac{OFX_t}{PWM_{fuel,t}} \tag{A.21}$$

$$OFX_t = \sum_c shroe_c PWE_c QE_c + TR_{gov,row,t} + NFFG_t$$

$$d \ln QMBAR_{fuel} = d \ln \varphi + d \ln OFX - d \ln PWM_{fuel}$$

- QMBAR** import ceiling; **QM** official imports
- PRQMBAR** rent rate: tariff equivalent of the shortage
- YPRQMBART** quota rent, rebated to users
- φ** share of official FX to fuel: 0.355 (2019)
- OFX** official FX inflows, USD
- shroe·PWE·QE** surrendered export receipts
- TR gov,row** grants; **NFFG** government net foreign borrowing
- $\perp$ either the ceiling binds with a positive rent, or it is slack with zero rent

Reading. Without any policy change, a fall in exports or aid, or a rise in the world fuel price, tightens the pump quota one-for-one in logs. Observed years use BRB's volumes, and the φ they imply is carried forward.

Smuggled fuel caps the market price once the shortage is deep

$$PM_t^{inf} = PWM_{fuel} (1 + \mu) EXR PREXR \left(1 + \frac{QMI}{QM_{fuel}^0}\right)^{1/\eta} + \text{margins} \tag{A.22}$$

$$QMI \geq 0 \perp PM_{fuel} \leq PM_t^{inf} \tag{A.23}$$

- QMI** informal fuel volume (DRC and Tanzania)

μ = 0.9 markup over the world price; of which μ ab = 0.1 transport paid abroad

QM 0 2019 official fuel imports
- PM inf** landed informal price

η = 2 supply elasticity of informal fuel

PM fuel market price: pump price plus quota rent

Reading. Informal fuel is always dearer than the pump: it is bought with parallel-rate FX plus a markup. It flows only when the quota rent lifts the market price to its cost, and from then on it caps that price. Only PWM(1 + μ ab) leaves the country; the premium and the smugglers' margin are domestic rents. μ is calibrated to 2024 black-market prices: 3.34 × pump against 3.35 observed.

Linear expenditure demand, with a fuel-to-firewood switch

$$SAV_i = \bar{\alpha}_i \text{CPI} + MPS_i (1 - ty_i) YI_i, \quad EH_h = (1 - ty_h) YI_h - SAV_h - \sum_i TRII_{i,h} \tag{A.24}$$

$$PQD_{c,h} QH_{c,h} = PQD_{c,h} \gamma_{c,h} POP_h + \beta_{c,h} \left(EH_h - \sum_{c'} PQD_{c',h} \gamma_{c',h} POP_h \right) \tag{A.25}$$

$$HXI_h = \left[\frac{PQD_{fuel,h} / PQD_{fuel,h}^0}{PQD_{agr,h} / PQD_{agr,h}^0} \right]^{-\varepsilon} \tag{A.26}$$

YI income: factors, transfers, rents	SAV, MPS, $\bar{\alpha}$ saving, propensity, indexed constant	ty direct tax rate; TRII transfers paid
EH consumption spending	QH, PQD quantity demanded, purchaser price	y subsistence quantity per person; POP population
β marginal budget shares: income elasticities 0.84–1.64, Frisch –2.5	HXI share of the LES fuel budget spent on fuel (added)	$\varepsilon = 0.1$ switching elasticity; the rest buys firewood, charcoal (agr)

Revenue, spending rules, deficit financing, debt

$$YG = \text{taxes}(ty, tq, ta, tm, te) + EXR \cdot TR_{gov, row} + YF_{gov} + \text{rents}_{gov}$$

$$ty_i = ty_{i,t}^b (1 + ty01_i TYSCAL_t), \quad QG_c = qg_{c,t}^b (1 + qgc01_c QGSCAL_t) \tag{A.27}$$

$$EG + INVVALG - YG = EXR \cdot NFFG + NDFG \tag{A.28}$$

$$GDEBT_t = GDEBT_{t-1} + NDFG_{t-1} + r^d GDEBT_{t-1}$$

$$FDEBT_t = FDEBT_{t-1} + NFFG_{t-1} + r^f FDEBT_{t-1}$$

YG, EG revenue; current spending (consumption, transfers, subsidies)

ty b, qg b base paths of tax rates, real consumption

INVVALG public investment

TYSCAL, QGSCAL closure scalars: which one adjusts is a closure choice

NFFG, NDFG net foreign (USD) and domestic borrowing

GDEBT, FDEBT debt; $r^d = 3\%$, $r^f = 3.5\%$

Investment by account, financed at home and abroad

$$DKINS_{i,k} = dk_{i,k,t}^b ISCAL_k (1 + IADJ_i iadj01_k) + ddk_{i,k,t} \tag{A.29}$$

$$INVVAL = \sum_i SAV_i + EXR \cdot NFFINS - NDFG + rents_{inv} \tag{A.30}$$

$$\sum_c PWE_c QE_c + TR_{row} + SAVF = \sum_c PWM_c QM_c + PWM_{fuel}(1 + \mu^{ab}) QMI + PAY_{row} \tag{A.31}$$

$$SAVF = NFFG + NFFINS + INVVALF \tag{A.32}$$

DKINS real investment by account i
(government, private, FDI), capital type k
INVVAL private investment; **NFFINS** private
foreign financing

ISCAL, **IADJ** scalars; IADJ clears saving–
investment in scenarios
SAVF foreign saving; **INVVALF** FDI
ddk added public investment (infrastructure)
TR row, **PAY row** transfers and factor payments
with abroad

Recursive: capital, labour and debt link the years

$$QF_{k,a,t} = (1 - \delta_k) QF_{k,a,t-1} + DKA_{k,a,t-1} \quad (\text{A.33})$$

$$DKA_{k,a} = \left(\sum_i DKINS_{i,k} - T_k \right) \frac{QF_{k,a}}{\sum_{a'} QF_{k,a'}} \left[1 + \kappa \left(\frac{WFA_{k,a}}{WFA_k^{avg}} - 1 \right) \right] + T_{k,a} \quad (\text{A.34})$$

$$QA_a VXI_a = (\text{CES}) + \sum_k mpk_{a,k,t} (QFINS_{k,t} - QFINS_{k,t}^0) \quad (\text{A.6})$$

δ k depreciation: 5 % private, 2.5 % public

WFA avg average return on capital type k

DKA new capital by activity

mpk marginal product of public capital (65 % return; zero in the reference path)

$\kappa = 0.5$ response to relative returns

QFINS public capital stock; **T k,a** capital from FDI targeted to activity a (added; mining in combi-x)

Also between years: labour by type grows with demographics (working-age population ≈ 3.1 % a year, 2026–40); debts accumulate (block 9). Expectations are myopic: investment follows current returns.

Scenarios switch to a foreign-exchange-constrained closure

Block	Reference path	Scenarios
Numeraire	domestic producer prices	same
Government balance	direct-tax rate clears	direct-tax rate clears
Spending and financing items	shares of GDP	real values, fixed tax rates
Saving–investment	savings rate clears	investment clears
Rest of world	private financing clears to 2027; premium clears from 2028	real exchange rate clears; financing exogenous
Labour / capital / land	wage curve / sector-specific / sector-specific	same
Mining resource	full employment	idle-resource option (combi)

So gains in a scenario show up as **investment** and a **real depreciation**, not as a higher savings rate or more foreign borrowing.

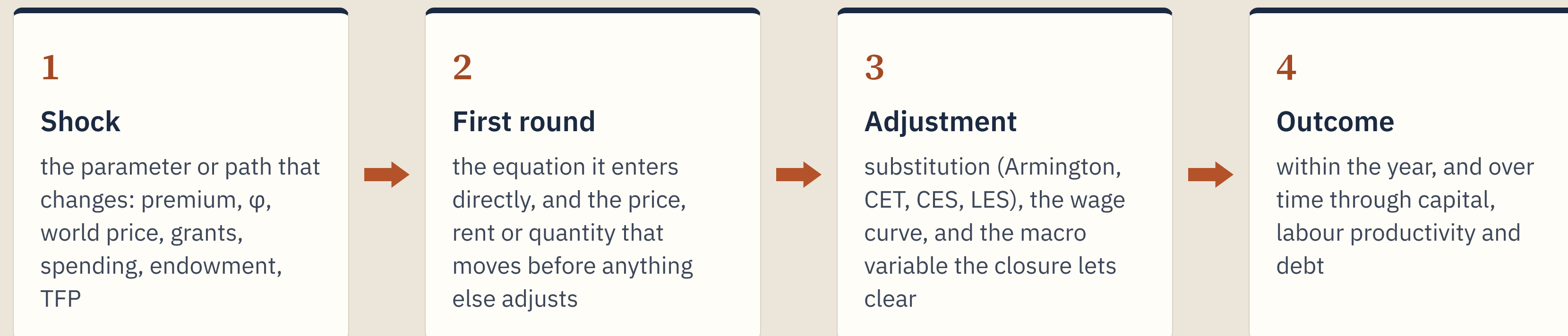
A square system each year, complementarities by sweep

1	2	3	4
Closure rules	Pairing	Newton	Sweep
fix the exogenous variables, read from the data	bipartite matching pairs 2,700 equations with 2,700 variables and checks the Jacobian is non-singular	exact derivatives (CasADi), year by year, warm starts and continuation for large shocks	quotas, idle mining resource and informal fuel: switch cells on the wrong branch, re-solve until none change

Two-pass calibration as in GEM-Core: pin GDP and back out TFP, then free GDP and re-base scenario parameters on the reference path.

3 · TRANSMISSION CHANNELS

Every channel is traced in four steps



Step 3 is where closure matters: under the scenario closure, the real exchange rate clears the external account and investment clears saving–investment; the government balance is cleared by the direct-tax rate.

3 · CHANNEL: EXCHANGE-RATE UNIFICATION (PREXR → 1)

Cheaper imports and the end of rent-seeking carry the gain

Channel	First round	Adjustment	Outcome
1 Import prices	PREXR m ↓ ⇒ PM ↓ on 71 % of imports (A.16, A.18)	QM/QD ↑ (A.14); cheaper intermediates and capital goods, PK ↓	real investment ↑ ⇒ capital ↑ (A.33)
2 Premium rent	YPREXRT → 0 (A.19)	transfer part: capital income ↓, prices ↓; cost part: rent-seeking resources freed, RW ↑ (A.5a)	TFP ≈ +5 % by 2040 at ω = 0.25, over ~3 years
3 Exporters	PE ↓ on ¼ of coffee and tea receipts (A.17)	CET shifts sales home (A.15)	first-round export loss
4 External balance	imports ↑ with SAVF fixed (A.31–A.32)	REXR ↑, about ×2 by 2040 (rowclos 1)	exports ↑ via CET (A.15)
5 Quotas lifted	QMBAR × 45 ⇒ slack, PRQMBAR = 0 (A.20)	fuel and chemicals cheaper	input costs ↓

ω = 0 (GEM-Core)

+1.05 pp · +16 %

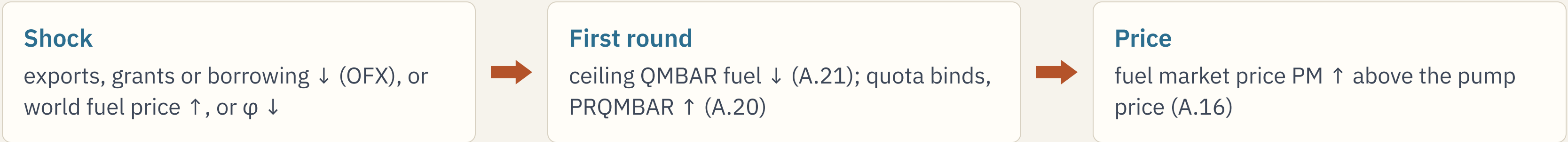
ω = 0.25 (default)

+1.49 pp · +23 %

ω = 0.5

+1.95 pp · +32 %

Less official FX becomes a pump shortage



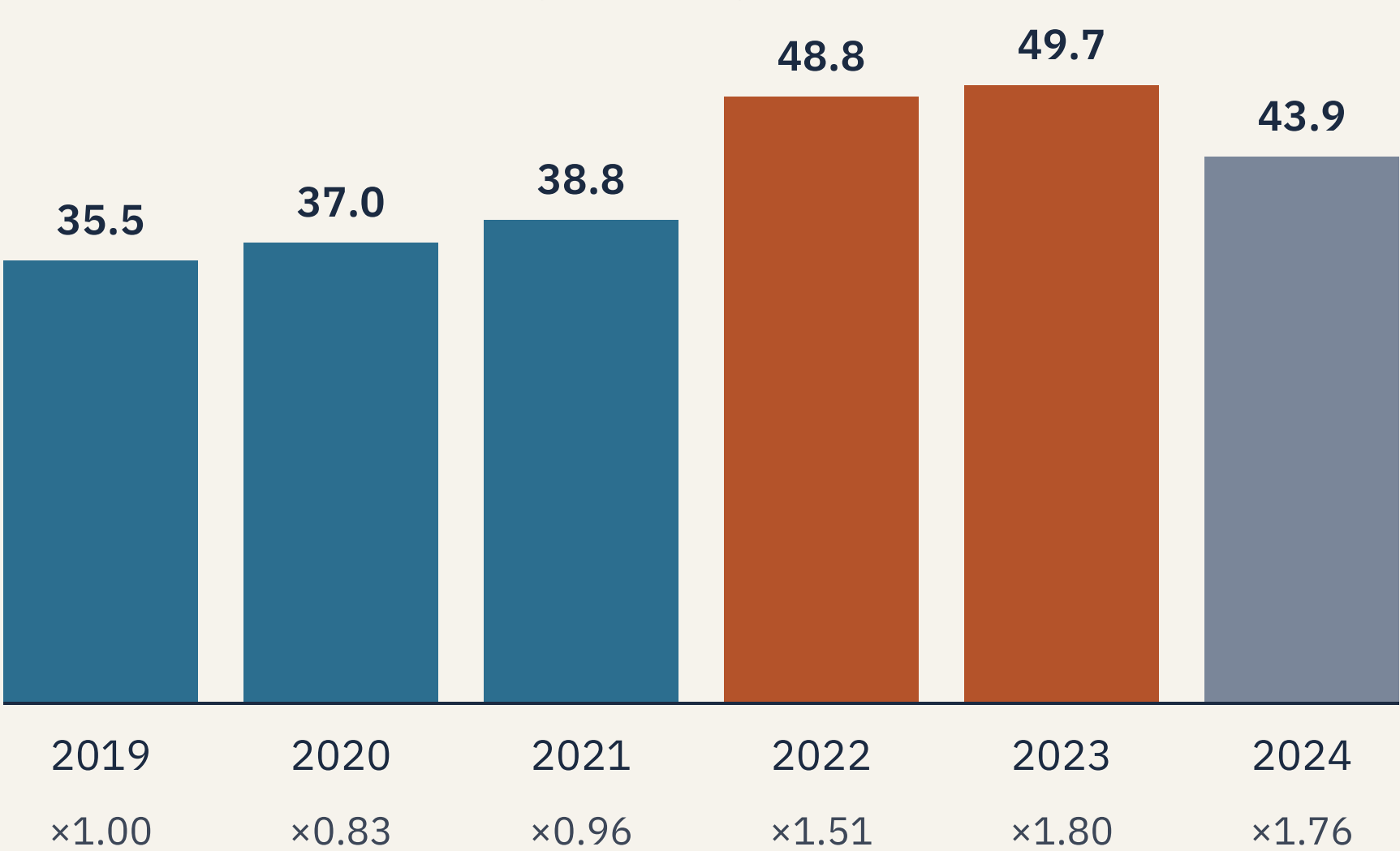
$$d \ln QMBAR_{fuel} = d \ln \varphi + d \ln OFX - d \ln PWM_{fuel}$$

Adjustment	Mechanism	Outcome
Firms	FXI ↓, VXI ↑ as fuel's relative price rises (A.9–A.10)	less output per worker in electricity, transport, construction
Households	HXI ↓: fuel budget shifts to firewood, charcoal (A.26)	fuel demand ↓; farm and forest demand ↑
Informal supply	once PM reaches PM_{inf} , $QMI > 0$ (A.22–A.23)	caps PM ; rents to smugglers; CA pays $PWM(1 + \mu_{ab})$

Holds under any closure: the ceiling is tied to official FX directly. Appendix A.5.2

BRB kept fuel flowing by giving it half of official FX

Fuel's share of official foreign exchange, %



Second row: world fuel price in USD, 2019 = 1

2022–23: world fuel prices rose ~80 %, official FX only ~20 %. BRB raised fuel's share from a third to a half.

2024: export receipts fell, the share fell to 44 %, official fuel fell 24 %. Model: 3.34 × pump against 3.35 observed.

Counterfactual at the 2019 share: shortage from 2020, crisis in 2022 (2.6 × pump).

Incidence depends on what finances the deficit

First round, any closure: QG ↑ (A.27) raises demand for public administration, education and health: labour-intensive, non-traded services ⇒ wages ↑, unemployment ↓ (A.12).

Financing closure	What clears (A.28)	Adjustment	Outcome
govclos 1 (default)	direct-tax rate, TYSCAL ↑	disposable income ↓ ⇒ consumption and saving ↓ (A.24)	investment ↓ via saving (investment clears)
govclos 2	domestic borrowing, NDFG ↑	private investment ↓ one-for-one (A.30); GDEBT ↑	crowding out; slower capital growth
govclos 3	foreign borrowing, NFFG ↑	SAVF ↑ (A.32) ⇒ REXR appreciates; OFX ↑ ⇒ fuel quota ↑ (A.21)	exports ↓; fuel supply ↑; FDEBT ↑

Grants cut both ways. A fall in grants lowers revenue (block 9) and official FX, so it tightens the budget and the fuel quota at the same time.

Demand now, productivity later

Public investment (inf)

Shock. ddk = 2 % of reference GDP: ramps up 2026–29, flat to 2035, out by 2040 (A.29)

Adjustment. public capital QFINS ↑ (A.33) ⇒ output of 12 activities ↑ through mpk, 65 % return (A.6)

Outcome. +0.53 pp GDP growth over uni

Extended (-x). roads cut transport costs 15 % and farm and food marketing margins 10 %; hydropower halves fuel per unit of electricity (ica, margins) ⇒ +1.4 pts to 2040 GDP at market prices

Human development (hd)

Shock. real education × 1.62, health × 1.85, phased in 2026–35 (A.27)

Adjustment. services demand ↑ now; from 2031 labour productivity FPRD ↑ with the human-capital-index gain (A.1)

Outcome. +0.28 pp GDP growth over uni+inf

Extended (-x). 2 % of the labour force moves from primary to secondary education by 2040 (labour in persons; secondary earns 3–10 × primary) ⇒ +1.6 pts to 2040 GDP

External shocks move income, the exchange rate and the fuel quota

Shock	First round	Adjustment	Outcome
Mining endowment × 3 (combi)	resource supply ↑; used as far as demand at the reference rent allows (A.13)	mining output ↑, 67 % exported; FX ↑ ⇒ real appreciation (rowclos 1)	+0.33 pp GDP; rents by ownership
+ mining FDI (combi-x)	FDI ↑ ≈ 0.5–0.7 % of GDP a year, all to mining capital (A.34)	mining capital × 3 by 2035; more of the resource used	+1.4 pts 2040 GDP; profits foreign-owned
World fuel price ↑	PM fuel ↑ (A.16) and QMBAR fuel ↓ (A.21)	shortage: substitution, switching, informal fuel	costs ↑ in fuel-intensive sectors
Coffee, tea price ↑	PE agr ↑ (A.17); OFX ↑	exports ↑ (A.15); fuel quota ↑ (A.21)	rural incomes ↑; fuel supply ↑
Grants ↓	YG ↓ and OFX ↓	tax ↑ (govclos 1); fuel quota ↓	fiscal and fuel squeeze together
Remittances ↑	household income YI ↑ (block 8)	more FX from transfers (A.31) ⇒ real appreciation	consumption ↑, tradables ↓

rowclos sets what absorbs an FX shock: real exchange rate (1), private financing (2), premium (4). Exports, grants and government borrowing also move the fuel ceiling.

3 • SUMMARY OF LEVERS

Each lever has a channel and a closure it depends on

Lever	Main channel	Depends most on
Fiscal spending	demand for services; financed by tax, domestic or foreign debt	government closure; domestic debt crowds out private investment
Public investment	public capital raises TFP in 12 activities (65 % return)	the assumed return; the financing closure
Human development	education and health demand now; labour productivity from 2031	size and lag of the productivity effect
Mining	resource endowment; 67 % of output exported; rents by ownership	idle-resource closure
Aid and exports	income, the external balance, and the fuel quota via official FX	rest-of-world closure
Trade policy	tariffs and quotas through import prices and rents	Armington elasticities
Climate, productivity	sector TFP; food prices through the LES	trade elasticities; income elasticities

4 • VALIDATION

Five layers, from the code to Burundi's data

1 • Transcription

Every equation holds at the 2019 SAM to 4.5×10^{-13} ; the system is square and non-singular each year

2 • Replication of GAMS

Base run within 0.09 pp a year; scenario increments match; unemployment by type to 2–3 decimals

3 • Backcast 2019–24

Observed inputs (INSBU, BRB), GDP pinned: a test of composition, prices and external accounts

4 • Fuel crisis 2024

Informal markup calibrated to black-market prices: $3.34 \times$ pump against 3.35

5 • Reproducibility

One command re-runs all 18 simulations, audits every solution and checks 41 findings: all reproduce, also from a fresh copy

4 · REPLICATION

GDP and consumption match; exports and investment run short

≤ 0.09 pp

gap on every aggregate's growth,
reference path 2026–40

≤ 0.02 pp

gap on each scenario layer's GDP
increment

2–3 dp

unemployment by labour type matches,
every scenario

With every addition switched off, the port reproduces the GEM-Core application to Burundi it was ported from. The one gap: under *uni*, exports and private investment grow about a fifth less than in the original. The exchange-rate path matches, so the gap is in the supply response. The reference application's results are for internal World Bank use, so only the fit is shown here.

Structure and the 2024 shocks fit; import growth does not

Average growth, % a year	Model	Actual
Real GDP (pinned)	1.9	1.9
Private consumption	1.2	2.4
Fixed investment	2.2	4.6
Exports	−2.5	−1.2
Imports	0.6	5.6
Agriculture / services VA	2.5 / 1.3	1.1 / 2.7
Imports, 2024 only	−11.1	−10.1
Fuel price / pump, 2024	3.34	3.35

Fits: every 2019 share within 2 points; the current account's direction; mining's collapses in 2020 and 2024; the 2024 import fall and fuel crisis.

Misses: the 2021 import jump (+13 % in tonnes) after an FX inflow. Outside fuel, FX scarcity is a price here, not a volume limit. INSBU's 2022–23 real imports conflict with BRB's tonnages.

Every finding reproduces, also from a fresh copy

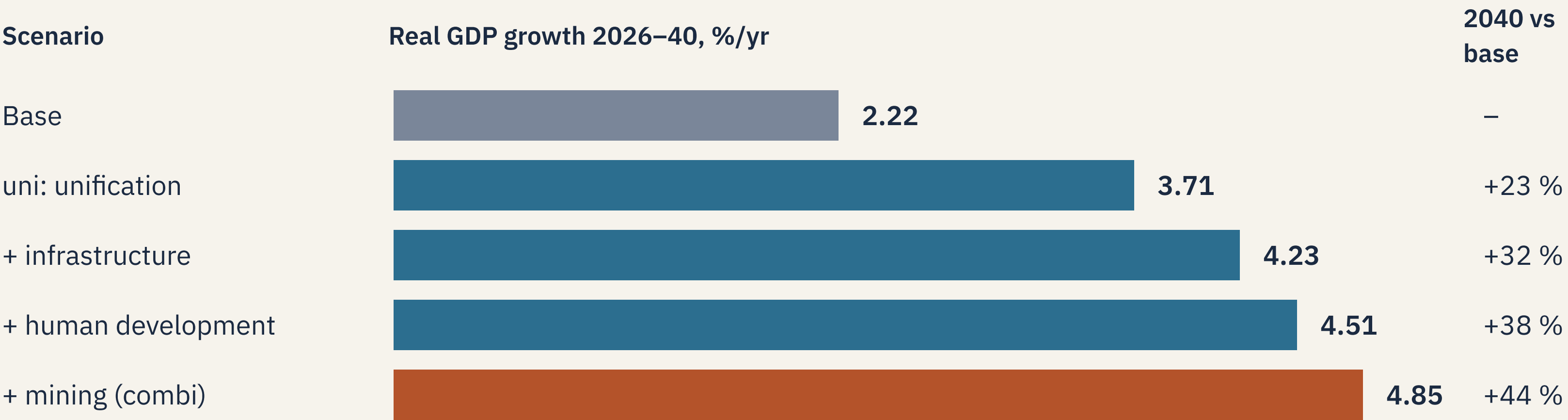
41 / 41 documented findings reproduced within tolerance	18 / 18 simulations converged	3×10^{-9} largest residual in the audit of every solution	8×10^{-13} largest gap between a fresh copy and the original
One command base-year check, all scenarios and backcasts, the audit, then 41 findings compared with Tables A.8–A.11 (0.005 pp on growth, 0.05 on levels)	Independent audit every equation at every year, Walras' law, quota, informal-fuel and idle-resource branches, lower bounds: no violation	Fresh copy package unzipped and set up with its own script, newer CasADi and NumPy: 13 minutes, identical results	

Five cumulative scenarios, from 2026

Scenario	What changes	Timing	Closure
base	no shock: GDP pinned at ≈ 2.22 %/yr, TFP backed out; fuel and chemicals quotas = 35.5 % and 40.2 % of official FX	2019–2040	financing clears to 2027, premium from 2028; savings rate clears
uni	premium halved, then set to 1; fuel and chemicals quotas $\times 45$ (slack)	2026 half; 2027 on full	real exchange rate and investment clear; real government spending
uni+inf	+ public investment of 2 % of GDP; public capital raises TFP in 12 activities (65 % return)	ramp 2026–30, full to 2035, out by 2040	as uni
uni+inf+hd	+ real education $\times 1.62$, health $\times 1.85$; labour productivity up to + 8.1 %	spending 2026–35; productivity 2031–40	as uni
combi	+ mining natural resource $\times 3$	ramp 2026–35	as uni, plus idle-resource closure
combi+	+ TFP gain reaching 17.5 % by 2040 (not yet implemented)	to 2040	as uni
uni+bs	uni + grants of 1 % of GDP (0.5 % in 2028), cutting taxes; - <i>inv</i> : invested instead	2026–28	as uni

All run with a quarter of the premium rent as a real cost ($\omega = 0.25$; block 5b). Extended -x variants add transport, marketing and hydropower effects to inf, the education mix to hd, and mining FDI to combi (slide 37).

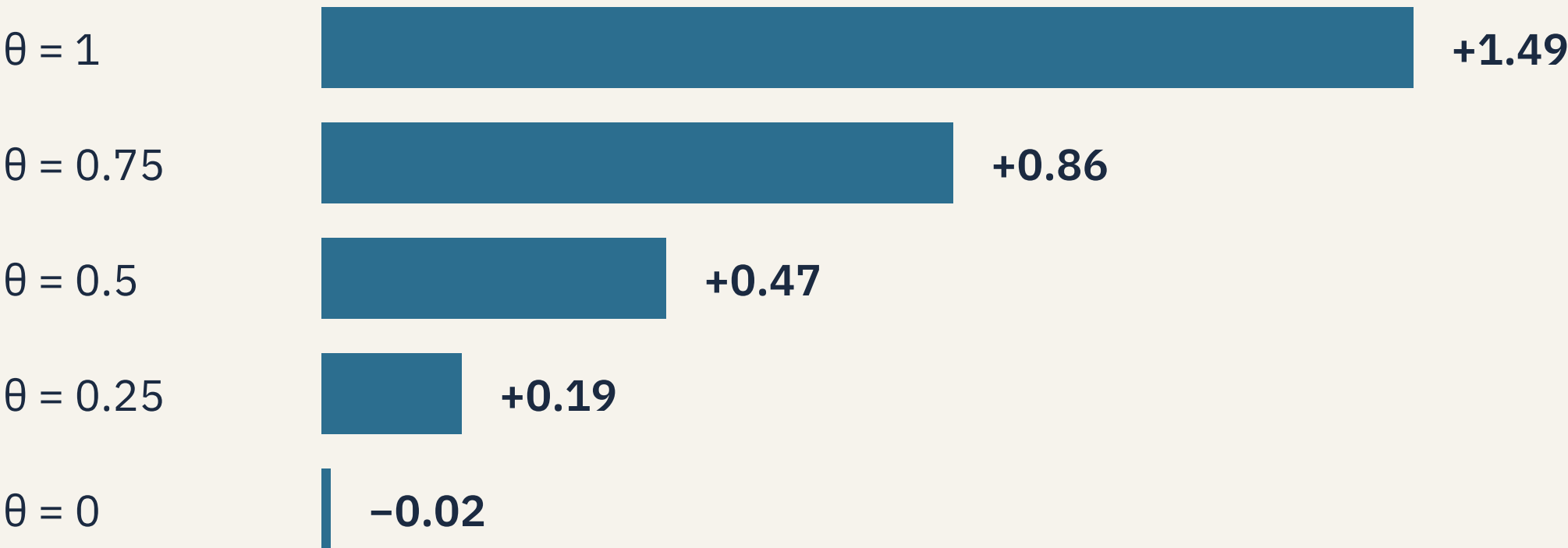
The full reform package adds 2.6 points a year



Full model, rent cost $\omega = 0.25$. Exports grow 7.6–9.9 % a year in the reform scenarios, against 3 % in the base; private consumption in 2040 is 27–46 % above base.

The unification gain scales with pass-through

GDP gain of *uni* over base, pp a year, 2026–40



Both main channels need pass-through: at $\theta = 0$ the premium on imports, and so the rent, does not change. At half pass-through the gain is under a third of the full figure.

2024 supports θ near 1: the premium rose 1.60 → 2.17 and imports fell 10 %, which only $\theta \approx 1$ reproduces.

Proposal: report +0.5 to +1.5 pp ($\theta = 0.5\text{--}1$).

Rent-seeking costs raise the gain; budget support cushions it

GDP in 2040 vs base	$\omega = 0$	$\omega = 0.25$	$\omega = 0.5$
uni	+15.9 %	+23.4 %	+32.0 %
uni + budget support	+15.9 %	+23.5 %	+32.0 %
... invested	+17.3 %	+24.8 %	+33.3 %
uni+inf	+25.0 %	+32.4 %	+40.9 %
uni+inf+hd	+30.1 %	+37.7 %	+46.4 %
combi	+36.2 %	+44.3 %	+53.5 %

Rent cost (ω). Each 0.25 of the rent treated as a real cost adds about 7–8 points to 2040 GDP in every scenario.

Budget support, 1 % of GDP in 2026–27 and 0.5 % in 2028: used to cut taxes, it lifts 2027 consumption by 2.5 points but leaves 2040 unchanged. Invested, it adds 1.4 points.

What the reforms build adds another 5 points by 2040

2040 level vs base, %	GDP, factor cost	GDP, market prices	Private consumption
uni+inf	+32.4	+36.2	+35.4
+ transport, marketing margins	+32.3	+37.4	+36.7
+ hydropower (uni+inf-x)	+32.5	+37.6	+37.0
uni+inf+hd → with education mix	+37.7 → +39.4	+40.5 → +43.5	+36.1 → +39.3
combi → combi-x (with mining FDI)	+44.3 → +47.4	+50.6 → +55.6	+45.9 → +50.8

Own increments, 2040 GDP at market prices: transport and marketing +1.2 · hydropower +0.2 · education mix +1.6 · mining FDI +2.0

combi-x grows 4.99 %/yr ($\omega = 0$: 4.61; $\omega = 0.5$: 5.41) against 2.22 % in the base.

Third group: rationing chemicals and fertiliser by official FX, like fuel, adds under 0.1 point. Rationing on the parallel market needs private-financing data.

Candidate simulations, runnable today

Question	How it is run
Gradual or partial unification	premium path over 2026–30, θ between 0.5 and 1
Fuel at the market rate	fuel bought at the parallel rate, quota removed: pump price to import parity
Oil price shock, +50 %	world fuel price; the quota tightens through official FX, informal fuel may switch on
Aid cut, –30 %	grants; fiscal and fuel-supply effects, under each government closure
Fuel or fertiliser?	reallocate official FX between the fuel and chemicals quotas
Coffee and tea boom	export prices raise official FX, and so the fuel quota
Paying for infrastructure	taxes, domestic debt (crowding out) or aid
Climate shock	agricultural TFP –10 %, one-off and recurrent

Six modelling choices the data do not settle

- 1 Official-rate imports are now FX-rationed (A.21). Should **the parallel market be quantity-rationed** too, and which data would identify it?
- 2 Is reporting unification as a **range over pass-through** θ (A.18), from 0.5 to 1, the right presentation?
- 3 How much of **the premium rent is a real cost** ($\omega = 0.25$, A.5a), and who receives the rest (A.19)?
- 4 **Base-year premium**: keep 2.5 for comparability, or re-base scenarios on the 2019 value of 1.58?
- 5 Is the reference rent the right **reservation price** in the idle-resource closure for mining (A.13)?
- 6 Stay on the **2019 SAM**, which allows a backcast, or move to INSBU's 2023 SAM, after the shocks?

NEXT STEPS

What would sharpen the model most

Data I am looking for

- BRB foreign-exchange sales by use: fuel, medicines, fertiliser
- Reserve drawdowns and import financing by source
- Informal fuel volumes: surveys or border data
- Black-market fuel prices for the first half of 2024

Model work

- Quantity rationing on the parallel FX market
- The last reference scenario (combi+)
- Close the export and investment gap under uni
- Household distribution beyond two groups

Thank you. Comments on any slide are welcome.

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Code, data, technical appendix and user guide: github.com/kodzovee9/burundi-cge